Arnav S. Nair

(609) 203-0694 | arnavnair229@gmail.com | linkedin.com/in/arnav-nair-005b82250 | arnavnair229.github.io

EDUCATION

Northeastern University - *Bachelor of Science in Business Administration (Dean's List)* **December 2026**
GPA: 3.9/4.0 **Major:** Mathematics & Finance; **Minor:** Data Science
Relevant Coursework: Stochastic Processes, Fourier Series & PDEs, C++, Applied Econometrics, Derivatives & Risk Management, Investments

PROFESSIONAL EXPERIENCE

Morgan Stanley **Starting January 2027**
Incoming Quantitative Analyst - Fixed Income Boston, MA

Morgan Stanley **January 2026 – Present**
Quantitative Researcher Intern - Fixed Income Boston, MA

- Built a Python Shiny order management tool for a newly launched SMA product after identifying gaps in Charles River's cross-asset (Corporate & Municipals) data and aggregate views; enables 3 PMs/Traders to evaluate trades across yield, duration/curve risk, cash impact, sector and state allocation, and tax-aware optimizations over 5 portfolios.
- Reduced RMSE by 5% on an ML-driven municipal bond bid-offer pricing engine by replacing AdaBoost with XGBoost and engineering new features; conducting ongoing research into hybrid deterministic and ML-based pricing layers to improve model robustness.
- Automated 20+ market risk and holdings reports for PMs/Traders via SQL-driven, Python, Tableau, and Excel pipelines.

KKR **June 2025 – August 2025**
Institutional Actuarial & Risk Modeling Intern New York, NY

- Automated analysis pipelines in Python and Streamlit to visualize surrender and mortality metrics in an interactive dashboard, improving processing time by 95%.
- Modeled future cash flows of life and annuity products with stochastic modeling in R3S, incorporating mortality, lapse, and surrender assumptions across 10M+ policyholders.
- Spearheaded a DuckDB-backed, object-oriented dashboard for 5+ reinsurance deals, improving efficiency by 70% and providing a scalable framework for future deals.

PROJECTS

Kalshi Quantitative Research & Systematic Trading **July 2026 – Present**
Co-developer, two-person team; built with AI coding agents

- Co-developed an end-to-end ML trading system across 9 Kalshi markets, covering the entire process from data and signal generation to order execution, engineered for real-world constraints (fills, data latency, order handling); walk-forward retraining on 3M intraday rows (5+ years, 10+ sources). Backtested +313% / +753% / +613% (2024 / 2025 / 2026 YTD) on a $2,000 account under adverse-fill stress; +314% for 2026 with realistic data latency.
- Authored 45 of 106 research reports and ran large-scale searches (24k trials; nine 2,000-trial Optuna studies; 13,600 walk-forward configs) under purged, embargoed CV; used deflated Sharpe, PBO, shuffle nulls, and pre-registered holdouts to reject the top config of a 24k-trial search.
- Wrote a 40-finding validation audit of the backtester and execution engine, making backtests reproducible and closing three silent-failure paths in order handling; identified data latency as the largest backtest-to-forward gap and rebuilt walk-forward on realistic timing.

HD Fund, LLC – Head Quantitative Analyst **March 2025 – October 2025**

- Researched, authored, and implemented a Kalman filter-based statistical arbitrage strategy, running robust backtests and achieving a 60% win rate while tuning entry and exit thresholds for 200+ trades over 3 years.
- Collaborated with fundamental analysts and quantitative developers to build a sentiment analysis ML model that produced revenue drivers for 15+ companies, improving accuracy of revenue forecasts by roughly 30%.

ACTIVITIES & LEADERSHIP

Northeastern University Systematic Alpha **January 2024 – May 2025**
Researcher & Developer Boston, MA

- Conducted Markov Chain and Monte Carlo simulations to design and validate an algorithmic trading model, enhancing the Sharpe ratio from 0.2 to 0.7.
- Achieved 22–39% APY over 4–10 years of back-testing using Markov Chains and options hedging strategies.

SKILLS & INTERESTS

Skills: Python (Polars, Pandas, Scikit-learn, NumPy, SciPy, Shiny, Streamlit, FastAPI), ML (XGBoost, Optuna, linear regression, clustering), C++, SQL, AWS, Snowflake, Tableau, LLM agents and harness design, AI-assisted development
Certifications: SOA Exam P (Score: 9), WSP: Excel, Accounting & Financial Statement Analysis, Bond & Math Analysis
Languages: English, Mandarin, Hindi **Interests:** Origami, Pool, Bio-Hacking, Skincare